ÖZYEĞİN UNIVERSITY

PHYS102/Fall 2017-2018 Final

SOLUTIONS

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This exam is out of 100 points. Contains 5 problems and 4 Consceptual questions.
- This exam is for 120 minutes.
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- **Answers without work shown or proper explanation will not be given any credit.**.

Questions	1	2	3	4	5	TOTAL

Formula Sheet

Coulomb's law:

$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2}$

$\frac{1}{4\pi\epsilon_0} = 9\times 10^9\ \mathrm{N\cdot m^2/C^2}$

Electric field:

$\vec{E} = \frac{\vec{F}_0}{q_0}$

$\epsilon_0 = 9\times 10^{-12}\ \mathrm{C^2/N\cdot m^2}$

Electric potential energy:

$W^E_{a\to b} = -\Delta U = U_a - U_b$

$U = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$

$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$

Electric flux:

$\Phi_E = \int E\cos\phi\, dA = \int \vec{E}\cdot d\vec{A}$

Gauss's law:

$\Phi_E = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{\mathrm{enc}}}{\epsilon_0}$

Electric potential:

$V = U/q_0$

$V = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r}$

$V_a - V_b = \int_a^b \vec{E}\cdot d\vec{\ell}$

$E_x = -\frac{\partial V}{\partial x},\ E_y = -\frac{\partial V}{\partial y},\ E_z = -\frac{\partial V}{\partial z}$

$W^E_{a\to b} = -\Delta U = -q\Delta V = -q(V_b - V_a)$

Capacitors and capacitance:

$C = Q/V_{ab}$

$C = \epsilon_0 A/d$

Capacitors in series:

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$

Capacitors in parallel:

$C_{eq} = C_1 + C_2 + C_3 + \dots$

(W^E is the work done by Electrostatic force.)

$\mu_0 = 4\pi\times 10^{-7}\frac{N}{A^2}$

Capacitors with a dielectric:

$C_\kappa = \kappa C_0\ ,\quad \kappa = \frac{\epsilon}{\epsilon_0}$

Energy in a capacitor:

$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$

Electric energy density:

$u = \frac{1}{2}\epsilon_0 E^2$

Current and current density:

$I = dQ/dt = n|q|v_d A$

$\vec{J} = nq\vec{v}_d$

Resistivity: $\rho = E/J$

Resistors: $V = IR,\quad R = \rho L/A$

Energy and power:

$P = (V_{\mathrm{a}} - V_{\mathrm{b}})I_{\mathrm{b}\to\mathrm{a}}$

$P = I^2R = V^2/R$ (for resistors)

Energy from power:

$U = \int P dt$ (if P is time dependent.)

$U = P(t_f - t_i)$ (if P is time indep.)

Resistors in series:

$R_{eq} = R_1 + R_2 + R_3 + \dots$

Resistors in parallel:

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

Kirchoff's Rules:

$\sum I = 0$ (junction rule)

$\sum V = 0$ (loop rule)

Charging RC Circuit:

<u>if $Q_0 = 0$:</u> $Q(t) = Q_f(1 - e^{-t/RC})$

<u>if $Q_0 \neq 0$:</u>

$Q(t) = Q_0 + (Q_f - Q_0)(1 - e^{-t/RC})$

$I(t) = I_0 e^{-t/RC},\quad I(t) = \frac{dQ(t)}{dt}$

Discharging capacitor:

$Q(t) = Q_0 e^{-t/RC}$

$I(t) = I_0 e^{-t/RC}$

Time Constant of an RC Circuit:

$\tau = RC$

Magnetic force on a moving charged particle:

$\vec{F}_B = q\vec{v}\times\vec{B}$

Magnetic force on a current carrying conductor:

$\vec{F}_B = I\vec{\ell}\times\vec{B}$

$d\vec{F}_B = Id\vec{\ell}\times\vec{B}$ (on a wire segment.)

Magnetic moment: $\vec{\mu} = I\ \vec{A}$

Magnetic torque: $\vec{\tau} = \vec{\mu}\times\vec{B}$

Potential Energy stored in a current carrying loop placed in B:

$U = -\vec{\mu}.\vec{B}$

Magnetic field of a moving charge:

$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$

Magnetic field of a current-carrying conductor: (Biot-Savart Law)

$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{\ell}\times\hat{r}}{r^2}$

Magnetic field of a long, straight, current-carrying conductor:

$B = \frac{\mu_0 I}{2\pi R}$

Magnetic force between current-carrying conductors:

$\frac{F}{L} = \frac{\mu_0 II'}{2\pi R}$

Magnetic fLux:

$\Phi_B = \int \vec{B}\cdot d\vec{A}$

$\Phi_B = \vec{B}\cdot\vec{A}$ (for uniform $\vec{B}$)

Faraday's Law: Induced EMF

$\varepsilon = -\frac{d\Phi_B}{dt}$

Motional EMF

$\varepsilon = \int \vec{v}\times\vec{B}\cdot d\vec{\ell}$

Ampere's Law:

$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 I_{\mathrm{enc}}$

$I_{\mathrm{enc}} = \oint \vec{J}\cdot d\vec{A}, (I = \vec{J}\cdot\vec{A},$ if J is constant)

Conceptual Questions (5 pts. each):

Q1: In electrostatic conditions all metal objects are equipotential surfaces, explain this statement using electric potential formulas.

Equipotential means: Same potential, i.e. no work done is needed to move the charges between equipotential points. And potential difference betweeen two equipotential points is ZERO.

$V(a) - V(b) = \int_a^b \vec{E} \cdot d\hat{l}$. Inside the conductor, E is zero and on the surface it is perpendicular to the surface. Thus $V(a) - V(b)$ is always zero for all points inside or on the surface of the conductor.

Q2: What is the essential feature for a charged particle to create a magnetic field? Does a charged particle always create both electric and magnetic fields, explain?

Magnetic field due to a charge: $\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$. Thus to create a magnetic field, any charge should have velocity, i.e. should move.

Q3: A current carrying conducting loop may experience a magnetic torque due to its magnetic moment $\vec{\mu}$. Explain how to find the direction of magnetic moment?

Magnetic moment: $\vec{\mu} = I\ \vec{A}$, thus it is parallel to the Area vector, i.e. normal vector. It is direction is determined by the right hand rule: Place your four fingers in the direction of the current in the loop, your thumb gives the direction of the magnetic moment.

Q4: A steady magnetic field is pointing into the page direction. In this region lies a circular conducting loop with its plane parallel to the page. If the radius of this circle is gradually decreasing find the direction of the induced current around the loop (clockwise or counter-clockwise), explain using either Faraday's Law or Lenz's Law (**not both**), and denote your choice in the explanation.

When r decreases, Area decreases, thus the flux decreases and $d\phi/dt < 0$.

Faradays Law: $\xi = -d\phi/dt$. Since $d\phi/dt$ is negative, ξ should be in positive direction. And positive direction is clockwise (when area vector is parallel to B).

Lens Law: Lenz's law states that the current induced in a circuit due to a change or a motion in a magnetic field is so directed as to oppose the change in flux and to exert a mechanical force opposing the motion.

Since decreasing radius causes flux to decrease, the induced current should result in a force in RADIALLY OUTWARD directions, on the loop, to increase the area. Thus the induced current, and emf should be in clockwise direction.

OR: decreasing area is analogous to decreasing B, thus induced current should create a B field in the same direction with the original B. Hence induced emf is clockwise.

Problem 1 (20 points)

A point particle with a positive charge $\boldsymbol{Q}$ is at the center of a non-conducting spherical shell with inner radius a and outer radius b. The charge distribution on the shell is not uniform but radially symmetric and the volume charge density is given as follows;

$$\rho(r) = \begin{cases} -\dfrac{\beta}{r^2} & a < r < b \\ 0 & \text{everywhere else} \end{cases}$$

To answer the rest of the question, we need to know the charge enclosed in the spherical shell with radius r (Gaussian surface), where $a < r < b$.

$Q(r) = Q + \text{charge stored from a to r} = Q + \int_a^r \rho(r) dV = Q + \int_a^r -\frac{\beta}{r'^2} 4\pi r'^2 dr' = Q - 4\pi\beta(r-a)$.

(a) (7 pts) Find the electric flux through a spherical surface with radius r in the region $a < r < b$.

$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$. Thus $\Phi_E = \frac{Q - 4\pi\beta(r-a)}{\epsilon_0}$.

(b) (7 pts) Find the electric field at a distance r from the center where $a < r < b$.

From Gauss Law $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$, and $\oint \vec{E} \cdot d\vec{A} = E 4\pi r^2$.
Thus $E(a < r < b) = \left(\frac{Q}{4\pi\epsilon_0 r^2} - \frac{\beta}{\epsilon_0} \left(\frac{1}{r} - \frac{a}{r^2} \right) \right)$.

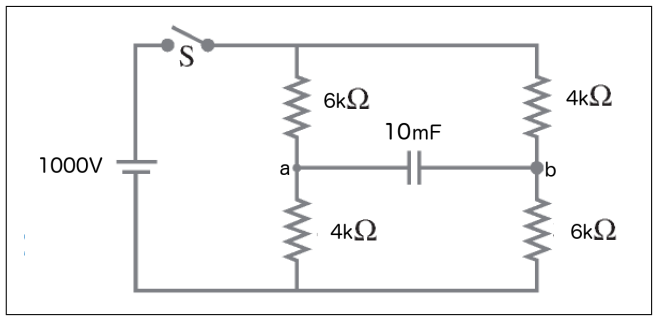
(c) (6 pts) Find the potential difference between the point in part (b) and the outer surface of the shell, i.e $V_r - V_b$.

$V_r - V_b = \int_r^b \vec{E} \cdot d\hat{l} = \int_r^b \frac{Q}{4\pi\epsilon_0 r^2} dr - \frac{\beta}{\epsilon_0} \int_r^b \frac{dr}{r} + \frac{a\beta}{\epsilon_0} \int_r^b \frac{dr}{r^2} = (\frac{Q}{4\pi\epsilon_0} + \frac{a\beta}{\epsilon_0})(1/r - 1/b) - \frac{\beta}{\epsilon_0} \ln(b/r)$.

Problem 2 (15 points)

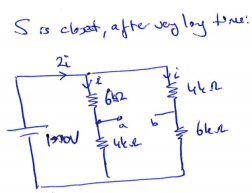
At the circuit shown, switch S is closed for a very long time so that the capacitor is fully charged.

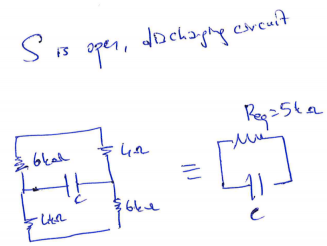
Suddenly, switch is opened at t=0 seconds. After the switch is opened,



(a) **(10pts.)** find the charge on the capacitor as a function of time.

The capacitor is totally charged with the potential V_{ba}, not 1000V ! We need to find V_{ba}, when the capacitors are totally charged. When the capacitors are totally charged, no current passes through, and the equivalent circuit is (left):





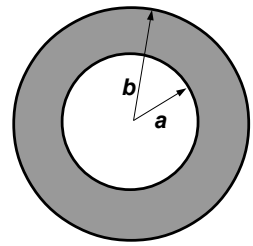
According to equivalent circuit (left), $i = 0.1$ A, and $V_b + i4k\Omega - i6k\Omega = V_a$ gives $V_{ba} = 200$ Volts. Thus, the capacitor is totally charged up to $Q_0 = VC = 2$ Coulombs.

When the switch is opened, capacitor is discharging and the equivalent circuit has $R_eq = 5k\Omega$ as shown (right). For the discharging circuit $Q(t) = Q_0 e^{-t/RC} = 2e^{-t/50s}$ C.

(b) **(5pts.)** find at what time (t') the charge on the capacitor becomes 0.1 C. (take $1/e = 0.37$, $1/e^2 = 0.14$, $1/e^3 = 0.05$, $1/e^4 = 0.02$)

$Q(t') = 2e^{-t'/50s}C = 0.1C$, and $\frac{0.1}{2} = \frac{1}{20} = 0.05 = e^{-3} = e^{-t'/50s}$. Thus $t' = 150$ s.

Problem 3 (15 points)



A cross-sectional view of a long, coaxial cable with inner radius $a = 1$ cm and outer radius $b = 3$ cm is given in the figure. Cable carries a uniform current of $I = 48$ A, into the page direction. (take $\pi = 3$.)

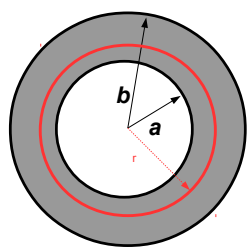
(a) **(5 pts.)** Find the magnitude of the current density.

Current density: $J = I/\text{area}$, since I is uniform.

Thus $J = I/\pi(b^2 - a^2) = 48A/3(3^2 - 1^2) \times 10^{-4}m^2 = 2 \times 10^4 A/m^2$, or $2A/cm^2$.

(b) **(10 pts.)** Determine the magnitude of the magnetic field at a point 2 cm away from the central line. (Take $\mu_0 = 1.2 \times 10^{-6}\text{T} \cdot \text{m/A}$).

According to Amphere's Law: $\oint \vec{B} \cdot d\hat{l} = \mu_0 I_{\text{enc}}$, and $I_{enc} = JA_{enc} = J\pi(r^2 - a^2) = 18A$.



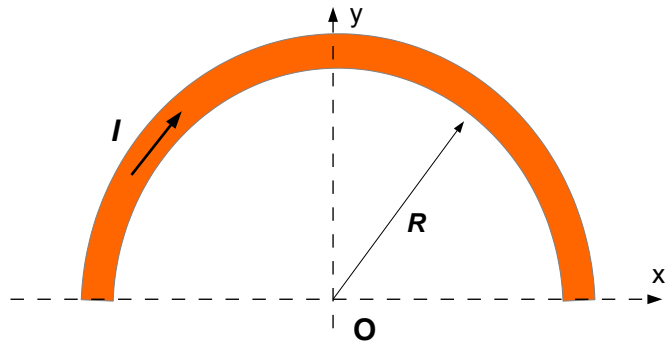
After drawing the Amphere's loop as shown, $\oint \vec{B} \cdot d\hat{l} = B2\pi r$.

Thus $B(r = 2cm) = \mu_0 I_{enc}/2\pi r = 1.2 \times 10^{-6} \times 18/(2 \times 3 \times 2 \times 10^{-2}) = 1.8 \times 10^{-4}\ T$.

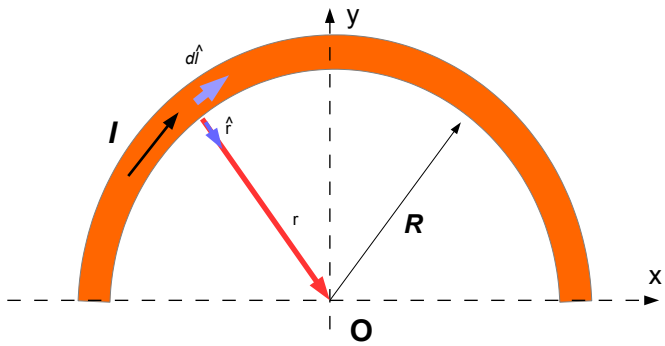
Problem 4 (10 points)

The semicircular wire segment with radius R is carrying current I in clockwise direction, and is placed in the $x-y$ plane as shown.

Using Biot-Savart's Law, find the magnitude and the direction of the magnetic field at the origin.



Biot-Savart Law: $d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{\ell}\times\hat{r}}{r^2}$, and the directions of the vectors are shown.



$d\vec{\ell}\times\hat{r} = dl\ 1\sin\pi/2$ (into the page) $= dl$ (into the page) $= Rd\theta$ (into the page). Thus:

$B(O) = \int_0^\pi \frac{\mu_0}{4\pi}\frac{IRd\theta}{R^2}$ (into the page) $= \frac{\mu_0 I}{4R}$ (into the page).

Problem 5 (20 points)

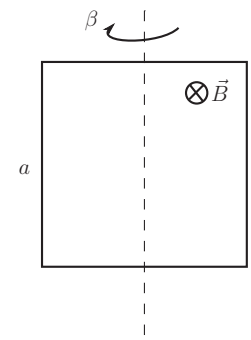
A conducting square loop with side a lies perpendicular to a uniform sinusoidal magnetic field $B = B_0 \sin \beta t$ into the page direction, where B_0 is a positive constant.

(a) (10 pts) Find the instant at which induced current on the square loop reverses direction **<u>for the first time</u>**. Is it from clockwise to counter-clockwise (CW $\rightarrow$ CCW) or from counter-clockwise to clockwise (CCW $\rightarrow$ CW)?

Flux: $\Phi_B = \vec{B} \cdot \vec{A} = a^2 B_0 \sin(\beta t)$
Induced emf: $\epsilon_{ind} = -\dfrac{d\Phi_B}{dt} = -a^2 \beta B_0 \cos(\beta t)$.
When the induced current is changing sign (direction), it should be zero :
$\epsilon_{ind} = -\dfrac{d\Phi_B}{dt} = -a^2 \beta B_0 \cos(\beta t) = 0$
Solution: $\cos(\beta t) = 0$, thus $\beta t = \dfrac{n\pi}{2}$. For the first time $t_1 = \dfrac{\pi}{2\beta}$.

Until t_1, emf is negative, thus it is CCW and then change to CW. (CCW $\rightarrow$ CW).

(b) (10 pts) If at $t = 0$ the square loop also starts to rotate with an angular speed of β, at what instant the induced emf would be zero **<u>for the first time</u>**.



At any time the angle between the area vector and B is $\theta = \beta t$, thus:

Flux: $\Phi_B = \vec{B} \cdot \vec{A} = a^2 B_0 \sin(\beta t) \cos(\beta t) = \dfrac{a^2 B_0 \sin(2\beta t)}{2}$
Induced emf: $\epsilon_{ind} = -\dfrac{d\Phi_B}{dt} = -a^2 \beta B_0 \cos(2\beta t)$.
emf should be zero : $\epsilon_{ind} = -\dfrac{d\Phi_B}{dt} = -a^2 \beta B_0 \cos(2\beta t) = 0$
Solution: $\cos(2\beta t) = 0$, thus $2\beta t = \dfrac{\pi}{2}$. For the first time $t = \dfrac{\pi}{4\beta}$.